EE681 – Compact Modeling	Date:18-02-2026
Major Project	
Instructor: Yogesh S. Chauhan	*Due Date: see below*

Note:

1. Use ADS simulator inside ICCAP software for parameter extraction project.
2. Download BSIM4 manual from BSIM website and Accellera Verilog-A Language Reference Manual from web.
3. Get ICCAP manual and other material from TA. Run examples to get acquainted with ICCAP software.
4. Prepare a report with references in IEEE double column style. Add all the plots in the report. Comment on issues encountered and how they were resolved. Send your report, Verilog-A code and ICCAP mdl file to TA.
5. Watch videos on Youtube to learn ICCAP software and Verilog-A coding.

Part-1 (Due date - 15-03-2026) - Extract BSIM4 model's parameters by fitting the TCAD data received from TA. Add plots in the report. Model must fit capacitance, current and derivatives accurately for all biases i.e.

- C_{GS}, C_{GD} and C_{GG} vs. V_{GS} at $V_{DS} = 0$.
- Subthreshold region for both low and high drain voltages.
- $I_{DS}V_{GS}$, g_mV_{GS} and g'_mV_{GS} in linear region for all gate voltages.
- $I_{DS}V_{GS}$, g_mV_{GS} and g'_mV_{GS} in saturation for all gate voltages.
- $I_{DS}V_{DS}$ and $g_{ds}V_{DS}$ for different drain and gate voltages.

Part-2 (Due date - 5-04-2026)
(i) *Implement* threshold voltage based MOSFET model for **(a)** terminal charge(s)/capacitance(s) including QME, and **(b)** drain-to-source current in Verilog-A including effects - Mobility degradation with vertical field, Velocity saturation, Channel Length Modulation, Drain Induced Barrier Lowering and Short Channel Effects.
(ii) Using your Verilog-A code, plot

(a) $I_{DS}V_{GS}$, g_mV_{GS} and g'_mV_{GS} for different V_{DS}.

(b) all possible normalized capacitances $\frac{C_{xy}}{W \cdot L \cdot C_{ox}}$ vs. V_{GS} for $V_{DS} = 0$, 50mV, $V_{DD}/2$, and V_{DD}.

(c) normalized capacitances $\frac{C_{GG}}{W \cdot L \cdot C_{ox}}$, $\frac{C_{GS}}{W \cdot L \cdot C_{ox}}$ and $\frac{C_{GS}}{W \cdot L \cdot C_{ox}}$ vs. V_{DS} for different V_{GS} values (0 to V_{DD}) in same plot.

* Model must **converge** for all extreme biases e.g. $\pm 100V$.
* Derivatives should be continuous and smooth up to second order for all biases including V_{ds}=0. Does your model work for V_{ds} <0?
* Check higher order derivatives.
* Check convergence on small circuits e.g. inverter, 15-stage ring oscillator etc.
* Run Gummel Symmetry Test.